

RVHS JC2 H2 Physics Preliminary Examinations Paper 3 Solutions

- 1 (a) Moment is the turning effect of a force.
The moment of a force is equal to the product of the force and the perpendicular distance of the line of action of the force from a pivot.

(b) (i) $\sum F_X = 0, Q_H = P$
 $\sum F_Y = 0, Q_V = W + M$
 $Q = \sqrt{Q_H^2 + Q_V^2}$
 $= \sqrt{(P^2 + (W + M)^2)}$ (shown)

(ii) Moment about P, $(M) \left(\frac{1}{4}\right) (3) + (W)(1.5) + (Q_H)(4) = (Q_V)(3)$
 $\left(\frac{3}{4}\right) (M) + (150) + (4200) = (3)(M + W)$
 $M = 1302 \sim 1300 \text{ N}$
Substituting back into expression from (b)(i),
 $Q = \sqrt{(920^2 + (500 + 1302)^2)} = 2023 \text{ N} \sim 2000 \text{ N}$

- 2 (a) By C O F